GENERIC POINTS OF THE SUPPORT AND EMBEDDED ASSOCIATED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The topological step (a closed subset of a locally Noetherian scheme is sober, so every point is a specialisation of a generic point) is the author's in-place argument that converts the cited inputs into assertions (1) and (2).

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The support of a finite type module is closed. Morphisms, Lemma morphisms-lemma-support-finite-type, used at proof lines 397-400. Makes Z = Supp(F) a closed subset.
- Generic points of the support are associated. Lemma lemma-minimal-support-in-ass, used at proof lines 400-402. Gives assertion (1) and the non-embedded half of (2).
- Associated points lie in the support. Lemma lemma-ass-support, used at proof lines 402-403. Places the associated points inside Z.

Source reference. Original lemma 05AD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6846–6846. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6850–6850. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 056J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6712–6712. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *the generic points of irreducible components of Supp($\mathcal{F}$) are associated points of $\mathcal{F}$, and*
(2) *an associated point of $\mathcal{F}$ is embedded if and only if it is not a generic point of an irreducible component of Supp($\mathcal{F}$).*

In particular an embedded point of X is an associated point of X which is not a generic point of an irreducible component of X.

Proof. Recall that in this case $Z = \mathrm{Supp}(\mathcal{F})$ is closed, see Morphisms, Lemma 056J and that the generic points of irreducible components of Z are associated points of $\mathcal{F}$, see Lemma 05AH. Finally, we have $\mathrm{Ass}(\mathcal{F}) \subset Z$, by Lemma 05AD. These results, combined with the fact that Z is a sober topological space and hence every point of Z is a specialization of a generic point of Z, imply (1) and (2). □